

## A Point-in-time foundation

**Market-data governance and target definition**  
Provenance · Asia/Bangkok cutoff · purged t+1 labels · train-only fitting

## B Three architecture-paired audit branches

### Numerical / denoising

Full TA (116)  
vs  
Full TA + causal VMD (122)

### Multimodal / LLM

Market-only  
vs predicted news  
+ falsification controls

### Regime / explainability

Causal Bull-Sideway-Bear  
+ train-only SHAP  
+ size-matched comparator

**Common fixed architecture panel**  
LSTM | CNN | LSTM-CNN | LSTM-Attention | LSTM-CNN-Attention

**Common SET50 next-day direction contract**  
Balanced accuracy as the primary endpoint

**Historical audit**  
2022-2025

**Source-contingent**  
partial-2026

**Frozen SET100**  
same-exchange transfer